Temporal TV-Programme Recommendation for SWEET.TV: A Reproducible Comparison of Content and Co-View Methods

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Abstract

This report studies Top-5 TV-programme recommendation for SWEET.TV using viewing logs and programme schedules while preventing temporal leakage. Views were matched to scheduled airings, a fixed 0.80 coverage threshold created implicit interactions, and popularity, content-based, association, and hybrid rankers were compared on weeks 11-20 for training and weeks 21-30 for validation. Association rules achieved the highest validation MAP@5 of 0.329997, and the manually uploaded designated submission obtained a Public Score of 0.37135 and a Private Score of 0.36730. The offline evidence supports association rules for this assignment, but it does not establish that the same method would be best for a production platform without an online experiment.

Index Terms - recommender systems, implicit feedback, association rules, temporal validation, MAP@5.

1. Introduction

The task is to recommend five TV programmes to each requested user. Viewing logs do not contain programme identifiers directly, so each interval must first be matched to the schedule. The evaluation is temporal: weeks 11-20 form training, weeks 21-30 form validation, and weeks 31-42 provide only the test schedule. Future viewing behaviour must not enter any feature or fit step.

The main difficulties are sparse programme labels and unequal user histories. Many schedule rows have `tv_show_id = 0`, while some users have too little labelled history for a reliable individual profile. As a concrete input-output example, a user's matched historical airings become five ranked nonzero programme IDs that are available in the target schedule.

Contributions

- A reproducible interval-join pipeline converts raw viewing logs into high-coverage implicit programme interactions.

- Four rankers are compared under one temporal split, candidate pool, repeat policy, fallback, and MAP@5 implementation; the final strategy is selected only on validation.

2. Related Work

Content-based recommendation forms user profiles from item attributes [1]. Association-rule mining estimates recurring item relationships through support and confidence [2], which motivates the one-day co-view baskets used here. Reciprocal-rank fusion offers a score-free way to combine ranked lists [3]; in this study it is treated as a tested alternative rather than assumed to improve either component.

Offline recommender evaluation requires an explicit task, candidate set, and metric protocol [4]. Matrix factorization can model implicit interactions and was considered as an alternative [5], but content and association were selected for implementation because the schedule provides rich metadata and the behavioural log provides direct co-view evidence. The Kaggle competition page defines the dataset and target-screen-time convention used in the interaction design [6].

3. Method

1. Join	2. Label	3. Fit	4. Rank	5. Evaluate / Submit
Same channel + positive interval overlap	Union coverage >= 0.80; drop id 0	Train-only interactions	Target-schedule candidates; shared fallback	Validation MAP@5; then refit weeks 11-30

Figure 1. Controlled temporal pipeline. Weeks 31-42 views are never used.

3.1 Interaction Construction

A view and an airing are matched on the same channel when their time intervals have positive overlap. Overlapping view fragments for one user-airing pair are merged before coverage is divided by the scheduled duration, preventing double counting. Rows with `tv_show_id = 0` are dropped because zero represents unlabelled airtime rather than an item.

Our operational decision classifies an airing as watched when the merged intervals cover at least 80% of its duration. The Kaggle Data page states, "Only TV shows with high screen time (>= 0.8) were taken into account" [6]. This supports alignment with target construction, but 0.80 is not presented as a universal viewing definition imposed by Kaggle. It was fixed before validation and was not tuned.

Matched pairs	Q25	Median	Q75	Q80	Q90	Q95	>= 0.80
3,014,129	0.043	0.195	0.656	0.809	1.000	1.000	611,760 (20.2964%)

The cutoff lies close to the empirical 80th percentile and retains only 20.2964% of matched pairs. It is strict but reduces the chance that short or accidental tune-ins become positive programme interactions.

3.2 Shared Evaluation Protocol

Component	Definition
Time split	Train weeks 11-20; validate weeks 21-30; test schedule weeks 31-42 only
Metric	AP@5 denominator = min(5, number of relevant programmes); macro MAP@5 over 1,238 eligible users
Candidates	All 6,636 nonzero programme IDs in the target schedule
Repeat policy	Previously watched programmes remain eligible for every method
Cold start	Training popularity within candidates; schedule frequency and programme ID break ties
Fairness control	Same users, candidates, repeat rule, fallback, and evaluator for every method

3.3 Models and Selection

Method	Scoring and validation-selected configuration
Popularity	Watched training-airing counts; required non-personalized baseline
Content	IDF-weighted cosine profile over category, genre, year, director, and actor; basic features + log-count history weighting
Association	One-day baskets; confidence; minimum pair support 5; history limit 50; recency decay 0.95
Hybrid	Reciprocal-rank fusion; content weight 0.25; rank constant 60

Every model fills five slots by its ranked score and then applies the same fallback if needed. Parameters were selected only on validation. After selection, the final association model was refit on weeks 11-30 and ranked weeks 31-42 candidates.

Reproducibility
CPython 3.12.14; Python standard library only; seed 20261025; macOS 15.7.7 arm64; 12 logical CPUs. Total pipeline runtime: 47.1944 s; recorded peak memory: 401.8125 MiB. Final association fit and rank: 1.3025 s; content: 3.1407 s; hybrid fusion overhead after component ranks: 0.0866 s. Eight unit tests passed, including interval-union, MAP@5, ranking, and submission checks.

4. Experiments

4.1 Validation Comparison

Method	MAP@5	Hit Rate@5	Mean Recall@5	Selected
Popularity	0.296127	0.885299	0.378689	No
Content	0.176451	0.646204	0.225067	No
Association	0.329997	0.897415	0.404887	Yes
Hybrid	0.318384	0.888530	0.401225	No

Association was selected by the predeclared rule: highest validation MAP@5, with a fixed simplicity order only for exact ties. It outperformed popularity and the hybrid. The hidden Kaggle scores were observed only after this decision and did not affect method or parameter selection.

4.2 Business-Oriented Diagnostics

Method	Catalogue coverage (programmes)	Repeat rate	Mean self-information (bits)
Popularity	5	0.608730	0.956709
Content	305	0.409048	4.664870
Association	110	0.706349	1.093430
Hybrid	151	0.734762	1.316831

Association provided the best accuracy but covered 110 programmes and repeated known programmes at a rate of about 0.706. Content covered 305 programmes, repeated less, and had higher self-information, but its MAP@5 was much lower. Because MAP@5 is the assignment's primary metric, association is the justified final choice. A production decision would also require diversity, novelty, satisfaction, and online evidence.

4.3 Heavy Viewer

User `10500740270951698446` was selected by a fixed rule: among validation-eligible users with positive training history, choose the maximum watched-airing count, breaking ties by the smallest numeric ID. The user had 904 training airings and 97 unique programmes.

Method	Ordered Top-5 (programme id, short title, category; * = validation hit)	Hits
Popularity	1 2400480 TSN [Info]*; 2 240081 Facts [Info]; 3 12001682 SpongeBob [Kids]*; 4 20088 Breakfast 1+1 [Ent.]; 5 2400508 Today [Info]	2
Content	1 12002107 Aladdin [Kids]; 2 12002564 Friendship Is Magic [Kids]; 3 12002720 Sonic Boom [Kids]; 4 12002888 Puss in Boots [Kids]; 5 12002904 Nexo Knights [Kids]	0
Association	1 12001682 SpongeBob [Kids]*; 2 2400480 TSN [Info]*; 3 240081 Facts [Info]; 4 12002856 Alvin [Kids]*; 5 12001734 Fairly OddParents [Kids]	3
Hybrid	1 2400467 Vikna-News [Info]; 2 12002355 Kung Fu Panda [Kids]*; 3 12001682 SpongeBob [Kids]*; 4 12002771 TMNT [Kids]; 5 2400480 TSN [Info]*	3
Actual validation	12001682 SpongeBob [Kids], 198 airings; 12002856 Alvin [Kids], 55; 12001697 Avatar [Kids], 43; 2400480 TSN [Info], 36; 12002355 Kung Fu Panda [Kids], 36	-

Association retrieved three validation-relevant programmes across children's content and news. Content recognized the broad children's category but all five similar cartoons missed the saved relevant set. The example suggests that co-view rules captured this user's repeated combination of interests more precisely than coarse metadata. This is a user-level observation, not a general causal claim.

4.4 Light Viewer

User 12453026253397466416 was selected by the same rule at the minimum watched-airing count. The user had one training airing and one unique programme, so a stable individual profile was not available.

Method	Ordered Top-5 (programme id, short title, category; * = validation hit)	Hits
Popularity	1 2400480 TSN [Info]*; 2 240081 Facts [Info]*; 3 12001682 SpongeBob [Kids]; 4 20088 Breakfast 1+1 [Ent.]*; 5 2400508 Today [Info]	3
Content	1 20088 Breakfast 1+1 [Ent.]*; 2 200337 Morning with Inter [Ent.]*; 3 200477 Weekend Breakfast [Ent.]; 4 400471 Zarobitchany [Ent.]; 5 2400508 Today [Info]	2
Association	1 2400480 TSN [Info]*; 2 240081 Facts [Info]*; 3 2400508 Today [Info]; 4 20088 Breakfast 1+1 [Ent.]*; 5 2400467 Vikna-News [Info]	3
Hybrid	1 20088 Breakfast 1+1 [Ent.]*; 2 2400508 Today [Info]; 3 2400467 Vikna-News [Info]; 4 2400480 TSN [Info]*; 5 240081 Facts [Info]*	3
Actual validation	240081 Facts [Info], 9 airings; 2400480 TSN [Info], 3; 5200303 Investigation Led [Educational], 3; 20088 Breakfast 1+1 [Ent.], 2; 200337 Morning with Inter [Ent.], 1	-

Popular news and morning programmes produced useful matches. Popularity, association, and hybrid each obtained three hits, while content obtained two. With only one training observation, the recommendation lists depend mainly on general popularity and the shared fallback rather than a well-supported personal pattern.

4.5 Verification and Designated Submission

Manual AP@5 verification
For user 36938861791891792, hits occurred at ranks 1 and 5. I recalculated (1/1 + 2/5) / min(5, 5) = (1.0 + 0.4) / 5 = 0.28, matching the saved program result.

Automated checks confirmed 1,260 users in template order, five distinct positive programme IDs per user, no zero, missing, duplicate, or extra-index values, and candidate membership in weeks 31-42. The designated file was `submission_association.csv`; SHA-256: `7f955ed10a352fc6a85a257c7a0d66e99fd5fce87a73d5f35e412951db14b0f2`.

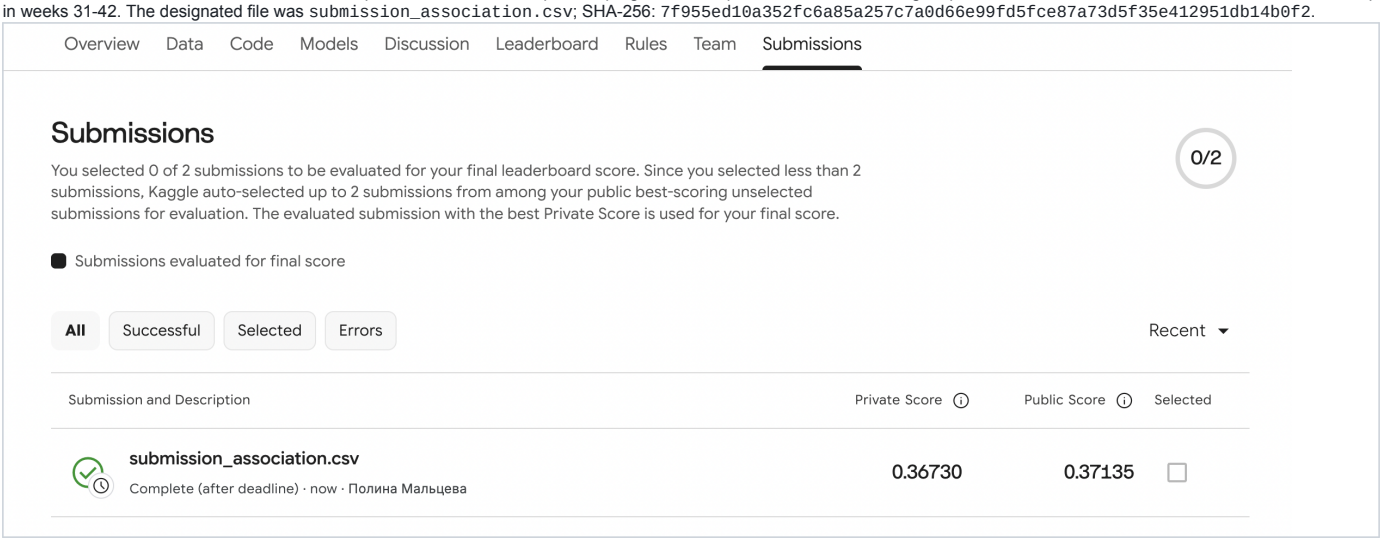


Figure 2. Kaggle Submissions page for the manually uploaded designated final file. Status: Complete (after deadline); Private Score = 0.36730; Public Score = 0.37135.

I selected the file in Finder, uploaded it manually, checked the filename, completion status, both returned scores, and saved this screenshot. No other submission was used to tune or replace the selected strategy.

5. Discussion

Final strategy: association rules
Validation MAP@5 was 0.329997 versus 0.296127 for popularity, 0.176451 for content, and 0.318384 for the hybrid. Popularity is weakly personalized; the hybrid is more complex but did not improve the selected metric. Association also ranked faster than content in the recorded final runs (1.3025 s versus 3.1407 s). I therefore select association for this offline assignment, while making no claim that it is necessarily best for a production platform without online testing.

5.1 Failure Case, Diagnosis, and Tested Change

Loop step	Observed evidence and interpretation
Detection	Content MAP@5 = 0.176451, below popularity; zero Top-5 hits for the heavy viewer
Root-cause hypothesis	Broad attributes made many children's programmes similar but did not encode the user's particular co-view pattern. This is plausible, not proven.
Tested change	Add association signal through reciprocal-rank fusion
Verification	Hybrid gave three heavy-user hits and MAP@5 = 0.318384, improving on content but remaining below association = 0.329997

5.2 Cold Start, Missing IDs, and Score Interpretation

Very short histories cannot support reliable individual profiles, so popularity fallback is less personalized but deterministic. The schedule contained 958,622, 952,889, and 1,150,604 zero-ID rows in the train, validation, and test periods; dropping them reduces the usable evidence to airings with identified programmes.

Validation MAP@5 was 0.329997; Public Score was 0.37135 and Private Score was 0.36730. The higher hidden-test values do not prove an improvement: the periods, users, and hidden targets differ. The scores were obtained after selection and were not used for tuning.

5.3 Reflection and Limitations

- What worked: one shared protocol made method numbers comparable; the interval union avoided double counting; association captured the heavy user's mixed children/news pattern.

- What surprised me: popularity remained strong for the light viewer; greater content coverage and novelty did not translate into higher MAP@5; hybridization improved content but diluted association.

- Next improvement: validate a switching rule that uses association only with sufficient history and a separately calibrated fallback for short histories.

The 0.80 threshold is an operational definition; viewing exposure is not enjoyment; absence from this platform log is not proof that a programme was never seen elsewhere. Offline metrics cannot establish effects on satisfaction, viewing, retention, or revenue without an online experiment.

6. AI Usage Disclosure

Tool: OpenAI Codex Desktop / CLI 0.158.0-alpha.2.1. Uses: requirement extraction, code generation and debugging, automated checks, CSV validation, evidence organization, reference verification, and report generation. Personally verified: AP@5 = 0.28 for user 36938861791891792; manual selection and upload of the designated file; filename, status, Public and Private Scores; saved screenshot. Not personally verified line by line: roughly nine million raw records, every join, all 6,300 submitted recommendations, every code line, runtime, memory, and SHA-256 internals. Those results rely on tests, validators, and saved evidence; Python runtime and operating-system reporting internals were not independently audited. The original, unmodified tool-native session export is submitted separately as the audit trail.

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